

Volume of Rectangular Prisms

Lesson 10-2

Name: _____

Date: _____

Class: _____

Key Vocabulary

Level 2 Standard

Picture first, then the word, then a plain-language meaning. Say each word out loud.



unit cubes

A box $2 \times 1.5 \times 1 = 3 \text{ ft}^3$ — it holds 3 cubic feet of stuff

Volume

Write the definition:



length $\times$ width $\times$ height

A tissue box, a fish tank, a brick — all rectangular prisms

Rectangular prism

Write the definition:



one cubic unit

$1 \text{ ft}^3 =$ a cube that is 1 foot on every edge

Cubic units

Write the definition:



A box with dimensions $4 \times 3 \times 2$ means $l = 4$, $w = 3$, $h = 2$

Dimensions

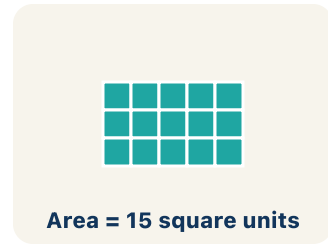
Write the definition:



*Cut a cereal box along its edges, unfold it flat —
that is the net*

Net

Write the definition:



*If the base is $6 \times 4 = 24 \text{ cm}^2$ and $h = 3$, then $V = 24$
 $\times 3 = 72 \text{ cm}^3$*

Base area

Write the definition:

Guided Notes

Level 2 Standard



WHAT WE'RE LEARNING TODAY

I can find the volume of a rectangular prism, including ones with fractional edge lengths, using base area $\times$ height.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Volume

Rectangular prism

Cubic units

Dimensions

Net

Base area

1 The amount of space inside a three-dimensional solid is its _____.

2 A solid with six rectangular faces is a _____.

3 The unit used to measure volume is _____.

4 The measurements of length, width, and height of a solid are its _____.

5 A flat pattern that folds into a solid is a _____.

6 The area of the bottom face of a prism, multiplied by height to find volume, is the _____.



Watch & Try — Worked Examples

See the notes in action: watch one worked all the way through, then try the next with the same steps.

 I do — watch

Follow each step as your teacher solves it.

Problem: What is the volume of a rectangular prism with $l = 4$ in, $w = 3$ in, $h = 5$ in?

- A. 60 in^3
- B. 12 in^3
- C. 60 in^2
- D. 24 in^3

Step 1 $V = l \times w \times h = 4 \times 3 \times 5 = 60$ cubic inches.

 **Answer:** A. 60 in^3

 We do — together

Solve this one with your class using the same steps.

Problem: A box has a volume of 36 cm^3 . Its length is 6 cm and width is 3 cm. What is the height?

- A. 2 cm
- B. 6 cm
- C. 3 cm
- D. 12 cm

Step 1 _____

Step 2 _____

Answer: _____

 You do — your turn

Now try one on your own. Show every step.

Problem: How is volume different from area?

- A. Volume measures 3D space (cubic units); area measures 2D surface (square units)
- B. Volume uses square units; area uses cubic units
- C. Volume only applies to cubes; area applies to all shapes
- D. There is no difference

Show your work:

Try It

Solve on your own. Check the answer key when you are done.

1. Two storage boxes are ready to pack. Box A is $5 \times 5 \times 2$ inches and Box B is $4 \times 4 \times 3$ inches. Which box holds more?

- A. Box A, because 50 cubic in is more than 48 cubic in
- B. Box B, because 48 cubic in is more than 50 cubic in
- C. They hold the same amount
- D. Box B, because it is taller

Show your work:

2. A keepsake box is 4 in long, 3 in wide, and $1\frac{1}{2}$ in tall. What is its volume?

- A. 18 cubic in
- B. 8.5 cubic in
- C. 12 cubic in
- D. 36 cubic in

Show your work:

Stretch Your Thinking

Level 2 enrichment

Challenge task — explain your reasoning in full sentences.

A toy company ships action figures in boxes that are $4 \times 3 \times 6$ inches. A shipping crate is $24 \times 12 \times 18$ inches. How many action figure boxes fit in the crate? Explain your reasoning step by step.

Sentence starter: Each toy box has a volume of ____ in^3 . The crate has a volume of ____ in^3 . I can fit ____ boxes because _____. I checked by _____.

Show your work:

Reflect — Exit Ticket

A rectangular prism has $l = 7$ ft, $w = 2$ ft, $h = 3$ ft. What is the volume?

- A. 42 ft^3
- B. 24 ft^3
- C. 42 ft^2
- D. 12 ft^3

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Volume
2. **Notes 2:** Rectangular prism
3. **Notes 3:** Cubic units
4. **Notes 4:** Dimensions
5. **Notes 5:** Net
6. **Notes 6:** Base area
7. **We Try (notes):** A. 2 cm — $V = l \times w \times h \rightarrow 36 = 6 \times 3 \times h \rightarrow 36 = 18h \rightarrow h = 2 \text{ cm}$.
8. **You Try (notes):** A. Volume measures 3D space (cubic units); area measures 2D surface (square units) — *Volume measures the space inside a 3D shape (cubic units like in^3). Area measures a flat 2D surface (square units like in^2).*
9. **Try It 1:** A. Box A, because 50 cubic in is more than 48 cubic in — *Box A: $5 \times 5 \times 2 = 50$ cubic in. Box B: $4 \times 4 \times 3 = 48$ cubic in. $50 > 48$, so Box A holds more.*
10. **Try It 2:** A. 18 cubic in — $V = 4 \times 3 \times 1.5 = 12 \times 1.5 = 18$ cubic in. *A fractional edge does not change the steps.*
11. **Exit Ticket:** A. 42 ft^3 — $V = 7 \times 2 \times 3 = 42$ cubic feet. *Remember: volume uses cubic units (ft^3).*